

2,2'-Bipyridine: An Efficient Ligand in the Cobalt-Catalyzed Synthesis of Organozinc Reagents from Aryl Chlorides and Sulfonates

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Abstract: The synthesis of functionalized arylzinc reagents from aromatic chlorides, triflates, or mesylates, via cobalt catalysis, is reported using 2,2'-bipyridine as a ligand in a mixture of acetonitrile and pyridine. This procedure allows the synthesis of a variety of functionalized arylzinc species in good to excellent yields. Some of these arylzinc compounds have been coupled with aromatic bromides under palladium catalysis.

Key words: cobalt halide, catalysis, zinc, aromatic chlorides, triflates

Although organozinc derivatives are widely used and well-documented reagents¹ their synthesis from functionalized aromatic chlorides has been rarely reported, and then mainly by electrochemical syntheses.² Their synthesis from aryl sulfonates has never been described to our knowledge. Indeed, these poorly reactive starting compounds are usually activated by solvated transition metal complexes such as Ni(0) or Pd(0) derivatives;³ magnesium or zinc, even when activated, insert only with difficulty into an aromatic carbon–chlorine or carbon–sulfonate bond. The price and availability of numerous functionalized aromatic chlorides or phenols, compared to the more reactive aromatic bromides or iodides, are strong driving forces towards the discovery of new activation processes.⁴

Over the past five years our research group has developed new cobalt-catalyzed processes for the activation of aromatic halides under mild conditions.^{2c–2f} Electrochemical methods have recently given way to convenient, entirely chemical processes.⁵ Indeed, functionalized aromatic bromides or iodides can be efficiently converted to the corresponding organozinc reagents in a single step. The reaction requires commercially available zinc dust and cobalt bromide, along with acetonitrile as solvent. However,

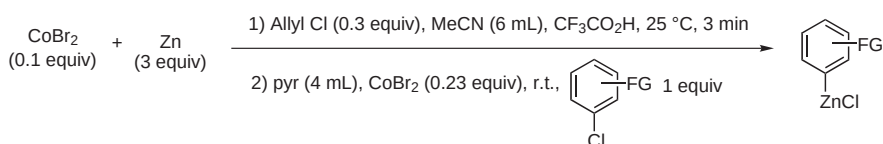
this process can not be employed to activate less reactive compounds such as aromatic chlorides. Electrochemical methods and electroanalytical studies have established that in situ generated low-oxidation-state cobalt(I) undergoes fast disproportionation in acetonitrile.^{6,7} However, the addition of pyridine allows the stabilization of Co(I), and thus, the activation of poorly reactive aromatic halides. The first entirely chemical application of these results was reported recently:⁸ the non-electrochemical synthesis of organozinc reagents from aromatic chlorides has been achieved by using cobalt catalysis and a mixture of acetonitrile and pyridine (Equation 1).

However, this method exhibits several drawbacks: the proportion of catalyst can reach as much as 33%, and reaction times can be as high as 24 hours, which hardly compares to the synthesis of organozinc reagents from aromatic bromides. Moreover, this last method does not allow the formation of arylzinc reagents from phenol derivatives.

Recent results in our group suggest that the replacement of CoBr₂ by CoBr₂(bpy) can improve these results.⁹ Indeed, addition of cobalt and 2,2'-bipyridine in a 1:1 ratio results in shorter reaction times in the case of aryl chlorides and allows arylzinc compounds to be synthesized from aryl triflates.

Thus we decided to add several ligands (PPh₃, dppe, and bpy) and cobalt, in both 2:1 and 1:1 ratios, to the reaction mixtures, which proved to be unsuccessful. However, 2,2'-bipyridine leads to faster reaction times when used in a 1:1 ratio with cobalt, as expected, whereas 2:1 mixtures with cobalt bromide retard the reactions.

From this point, several other parameters, including zinc dust and catalyst proportions, solvent composition, substrate concentration, and the implementation of a prelimi-



Equation 1

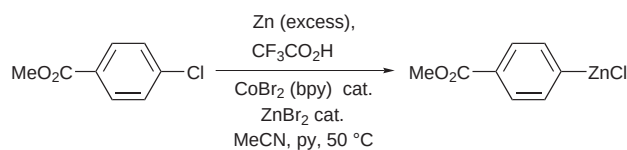
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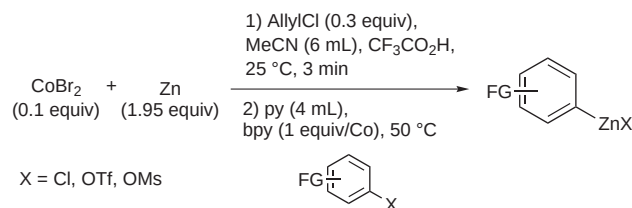
nary step⁵ have been investigated in order to optimize the reaction conditions with *p*-methyl 4-chlorobenzoate as the test substrate (Equation 2).



Equation 2

We found that MeCN–pyridine in a 6:4 ratio proved to be the best solvent mixture, as it leads to optimum reaction times and stopped further degradation of the organozinc reagent into ArAr. Carrying out the reaction at 50 °C instead of room temperature allowed us to reduce the amount of both Zn powder and cobalt catalyst to 1.5 and 0.1 equivalents, respectively. Moreover, if allyl chloride was added in a preliminary step – similarly to methods described in previous works⁵ – the amount of byproduct (ArH) formed decreased. The addition of allyl chloride also helped in suppressing the addition of hygroscopic ZnBr₂ as allyl chloride generates Zn²⁺ salts in situ during this preliminary step.⁵

The optimized conditions have been successfully applied to other poorly reactive aromatic reagents with sensitive functional groups (Equation 3).



Equation 3

The preliminary results indicate that this method is very sensitive to the substituents on the aromatic ring, as every single reagent gives rise to unique issues. As a consequence, the experimental conditions have to be adapted slightly for each case (Table 1).

Organozinc reagents are identified through conversion into their corresponding aromatic iodide by addition of iodine, followed by GC analysis. The standard conditions can be successfully applied to methyl 4-chlorobenzoate (Table 1, entry 1). In order to lower the reaction times, we have increased the proportion of zinc powder (Table 1, entries 2–4). The degradation of the organozinc reagent can additionally be achieved through increasing the amount of pyridine (Table 1, entry 3). It is possible to slow down the reactions by simply lowering both the amount of zinc powder and cobalt catalyst (Table 1, entry 5).

The synthesis of the organozinc reagent from 4-chlorobenzonitrile has revealed that, under the standard conditions (Table 1, entry 1), low-valent cobalt insertion takes place not only into the carbon–chlorine bond, but also into the carbon–nitrile bond, although, surprisingly this result had been observed before in rare cases, notably with nickel catalysts.¹⁰ This point has been circumvented by discarding the improved, 2,2'-bipyridine-containing catalyst, and returning to the classic cobalt bromide. A slight increase in the catalyst proportion was necessary in order to achieve satisfying reaction times (Table 1, entry 6).

Following these principles, we also looked at the synthesis of organozinc reagents starting from phenol derivatives^{11,12} (Table 2). Whereas the conversion of aryl triflates works fairly well (Table 2, entries 1–4), with few selectivity problems except in the case of 4-acetylphenyl trifluoromethanesulfonate (Table 2, entry 4), the less reactive aryl mesylates (Table 2, entry 5–8) give rise to preferential S–O bond cleavage, leading to the corresponding phenol. It should also be pointed out that neither phenyl acetate nor phenyl trifluoroacetate can be converted to the corresponding organozinc reagent under these conditions.

This is the first reported synthesis of organozinc triflates and proves that the methodology can be adapted to a wide range of aromatic reagents.

Various coupling reactions have been investigated by using arylzinc chlorides or triflates as synthesized according to the above-described procedure. These include reactions with allylic chlorides or acetates, and bromopyridine.

Table 1 Synthesis of Organozinc Chlorides under Optimized Conditions

Entry	ArCl	Catalyst (equiv)	Zn (equiv)	MeCN–py	Time (h)	ArH ^a	ArCl ^a	ArAr ^a	ArZnCl ^a
1	4-MeO ₂ CPhCl	CoBr ₂ (bpy) (0.1)	1.95	6:4	3	4	0	10	86
2	PhCl	CoBr ₂ (bpy) (0.1)	3	6:4	5 (18) ^b	3 (15) ^b	19 (32) ^b	15 (15) ^b	62 (38) ^b
3	4-F ₃ CPhCl	CoBr ₂ (bpy) (0.1)	3	4:6	3 (24) ^b	7 (7) ^b	4 (2) ^b	23 (57) ^b	66 (33) ^b
4	4-F ₃ CPhCl	CoBr ₂ (bpy) (0.1)	3	6:4	5 (24) ^b	6 (7) ^b	4 (2) ^b	30 (57) ^b	60 (33) ^b
5	4-MeCOPhCl	CoBr ₂ (bpy) (0.05)	1.7	6:4	1 (5) ^b	13 (20) ^b	0 (0) ^b	0 (6) ^b	87 (70) ^b
6	4-NCPhCl	CoBr ₂ (0.2)	1.95	6:4	1	3	0	8	89

^a GC yield of an iodinated aliquot.

^b In parentheses: GC yield under the standard conditions described in Equation 3.

Table 2 Synthesis of Organozinc Triflates and Mesylates

Entry	ArX	CoBr ₂ (bpy) (equiv)	Zn (equiv)	Time (h)	ArOH ^a	ArH ^a	ArX ^a	ArAr ^a	ArZnX ^a
1	PhOTf	0.1	3	8	–	6	7	0	87
2	4-EtO ₂ CPhOTf	0.1	1.95	2.5	–	1	4	19	77
3	4-NCPhOTf	0.1	1.95	0.5	–	0	0	19	81
4	4-MeCOPhOTf	0.1	1.95	1	29	4	0	7	61
5	PhOMs	0.1	3	72	–	–	100	–	–
6	4-EtO ₂ CPhOMs	0.2	1.95	21	74	8	17	0	1
7	4-NCPhOMs	0.1	3	24	78	3	0	0	19
8	4-MeCOPhOMs	0.1	3	8	13	51	0	0	36

^a GC yield of an iodinated aliquot.

The coupling of allyl chlorides or acetates gave a 35% yield (GC) of the coupled product, provided the coupling step is carried out with additional cobalt catalyst and 2 equivalents of the allylic reagent. However, the poor yield and large amounts of ArH and ArAr byproducts arising from ArZnCl or ArZnOTf prevented any further purification. The use of pyridine as co-solvent in the synthesis of the arylzinc reagents has been criticized for these poor results, but no clear evidence of its precise role has been reported so far. However, these reactions are in the early stages of development and future improvements may prove successful.

Some organozinc reagents mentioned in Tables 1 (entries 1, 4, and 5) and 2 (entry 8) obtained from aryl chlorides or triflates have been coupled with 2-bromopyridine (1 equiv) in the presence of a catalytic amount of PdCl₂(PPh₃)₂ (1%) at 50 °C. These well-known Negishi couplings¹³ proceeded in good yields exclusively if another solvent such as DMF or THF (5 mL) was added to the solution to improve solubility. Under these conditions, each 2-arylpyridine was isolated in 85% yield versus ArZnX; moreover, using a cobalt catalyst for carbon–carbon bond formation¹⁴ was not efficient, a palladium catalyst is essential.

An efficient and convenient cobalt-catalyzed synthesis of aromatic organozinc reagents in high yields has been developed with 2,2'-bipyridine as a ligand. This ligand is essential for the formation of arylzinc reagents from phenol derivatives.

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Phenol (5.3 g, 50 mmol) and pyridine (22 mL) were cooled to 0 °C in a three-way reaction vessel under an argon atmosphere. Tf₂O (8.5 mL, 50 mmol) was added dropwise over 45 min to the magnetically stirred solution, giving rise to a strong red-orange color. Then the reaction mixture was stirred at r.t. until the starting phenol had been consumed.

(monitored by GC on hydrolyzed aliquots). H₂O (40 mL) was added to the mixture, which was then extracted with CH₂Cl₂ (3 × 40 mL). The organic layer was treated with 10% HCl (40 mL), washed with a sat. solution of NaCl (40 mL), and dried over anhyd MgSO₄. After removing the solvent under vacuum, the product was purified by flash chromatography on silica gel (pentane) to give phenyl trifluoromethanesulfonate in 95% yield as a colorless oil. The same procedure was applied to the synthesis of mesylates, by replacing Tf₂O with MsCl.

Coupling of Organozinc Chlorides or Triflates with 2-Bromopyridine; Typical Procedure (Table 1, entry 1)

Preliminary Step: Zn powder (1.27 g, 1.95 equiv, <10 μm) and CoBr₂ (219 mg, 0.1 equiv) were placed in a reaction vessel under an argon atmosphere. MeCN (6 mL), allyl chloride (0.25 mL, 0.3 equiv), dodecane (0.2 mL, internal reference for GC), and TFA (0.05 mL) were added in one batch to the vessel. The resulting reaction mixture was magnetically stirred under a flow of argon at r.t. for 3 min. Synthesis of the Organozinc Reagent: Pyridine (4 mL), 2,2'-bipyridine (0.156 g, 0.1 equiv), and methyl 4-chlorobenzoate (1.71 g, 10 mmol) were added to the reaction mixture,

and stirred at 50 °C until the reaction was complete (aliquots were removed, monitored by GC after iodination). After 3 h, 4-carboxymethylzinc chloride was obtained in 86% yield (GC).

Coupling Reaction: DMF or THF (5 mL), 2-bromopyridine (1 mL, 1 equiv) and PdCl₂(PPh₃)₂ (70 mg, 0.01 equiv) were added to the reaction mixture obtained above, and the resulting solution was stirred for 3 h at 50 °C until 4-carboxymethylzinc chloride had been completely consumed. The solution was hydrolyzed with NH₄Cl (100 mL), extracted with Et₂O, the organic layer was washed with brine, dried, and the solvent evaporated. The residue was isolated by flash column chromatography (silica gel; pentane–Et₂O, 85:15) to give the Negishi coupled product in 73% yield. The structure was confirmed by ¹H and ¹³C NMR spectroscopy and MS.

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